



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Spectral Transient Shaper

Spectral Transient Shaper

Reshapes attack and sustain separately in every FFT bin, without crossover bands, then finishes the result with optional saturation, clipping, oversampling, and limiting.

1025 spectral bins

Simple or 3-band EQ detection

Attack, Sustain, and Delta monitoring

25 factory presets



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01 Overview

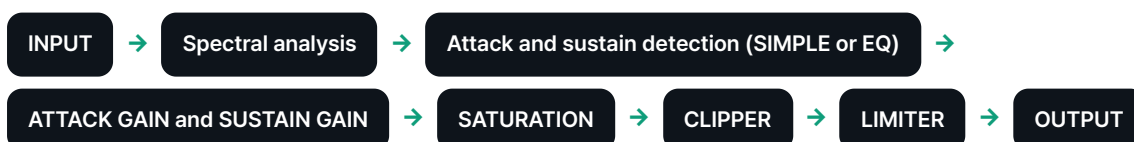
A conventional transient shaper moves the whole signal at once. TB Spectral Transient Shaper analyses the sound in 1025 frequency bins and decides, bin by bin, where transient energy should rise, where sustain should tighten, and where the sound should stay untouched.

ATTACK GAIN and **SUSTAIN GAIN** boost or cut the two branches, and each has its own release. SIMPLE mode uses one sensitivity across the spectrum; EQ mode gives the low, mid, and high regions their own sensitivity. Two analyzers show what is detected, the monitor lets you hear each branch or only the change, and an optional output chain adds saturation, a soft clipper, and a limiter.

Key features

- Attack and sustain shaping per FFT bin (1024-point FFT, 4× overlap), with no crossover bands
- **ATTACK GAIN** and **SUSTAIN GAIN** from –24 to +24 dB, each with its own **RELEASE**
- SIMPLE or EQ detection; EQ uses a low shelf, a bell, and a high shelf per branch
- ATTACK and SUSTAIN analyzers with the sensitivity curve and reference line
- Monitor NORMAL, ATTACK, or SUSTAIN, each with an optional DELTA
- Optional saturation (six characters), soft clipper, and lookahead limiter
- 1×, 2×, 4×, or 8× oversampling around saturation and clipping
- 25 factory presets in six groups, user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Spectral Transient Shaper 1.4.1 for Windows (VST3). The current macOS build is version 1.0.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Spectral Transient Shaper on a drum, bass, guitar, vocal, or bus track. By default **ATTACK GAIN** is +4 dB, **SUSTAIN GAIN** is –2 dB, and both releases are 50 ms.

- 2 Leave **SHAPER** 3 on SIMPLE and play the track. Watch the **ATTACK** 5 and **SUSTAIN** 6 analyzers.
- 3 Raise **ATTACK GAIN** 7 for more punch, or lower it to soften hits. Shorter **RELEASE** 8 makes tighter, sharper transients.
- 4 Lower **SUSTAIN GAIN** 9 to dry up room tails and ring, or raise it for more body.
- 5 Switch on **DELTA** 4 to hear only what the processing changes, then switch it off again.
- 6 Match the level with **OUTPUT** 15 and compare with **BYPASS** 2.

03 Interface



TB Spectral Transient Shaper 1.4 interface.

- 1 **Preset bar** — previous and next preset (hold to keep stepping), the preset browser, Save, Undo, and Redo.
- 2 **BYPASS** — bypasses all processing and reports zero latency while active.
- 3 **SHAPER** — SIMPLE or EQ detection.
- 4 **MONITOR and DELTA** — what you hear: NORMAL, ATTACK, or SUSTAIN, and only the difference with DELTA.
- 5 **ATTACK analyzer** — the attack branch's spectrum, sensitivity curve, and reference line.
- 6 **SUSTAIN analyzer** — the same for the sustain branch.
- 7 **ATTACK GAIN** — boost or cut of the detected attack.
- 8 **RELEASE (attack)** — how quickly attack shaping returns to neutral.
- 9 **SUSTAIN GAIN** — boost or cut of the sustained part.
- 10 **RELEASE (sustain)** — how long sustain shaping stays active.
- 11 **QUALITY** — oversampling x1 to x8 and the limiter LOOKAHEAD.
- 12 **SATURATION** — on/off lamp, character, Shaper Link, and DRIVE.
- 13 **CLIPPER** — on/off lamp and THRESHOLD.
- 14 **LIMITER** — on/off lamp and CEILING.
- 15 **GAIN** — INPUT and OUTPUT with level meters.

04 Controls

Shaper

Control	Range · default	What it does
ATTACK GAIN	-24 to +24 dB · +4 dB	How much the detected attack of every spectral band is boosted or cut. 0 dB is no change.
RELEASE (attack)	1 to 2000 ms · 50 ms	How quickly attack shaping returns to neutral. Short values make tighter, sharper transients.
SUSTAIN GAIN	-24 to +24 dB · -2 dB	How much the sustained part of every spectral band is boosted or cut. 0 dB is no change.
RELEASE (sustain)	1 to 2000 ms · 50 ms	How long sustain shaping remains active after detection falls away.
SHAPER	SIMPLE, EQ · SIMPLE	SIMPLE uses one sensitivity value across the spectrum. EQ uses low shelf, bell, and high shelf sensitivity bands.
Sensitivity	0 to 1 · 0.5	How readily each branch detects attack or sustain. Shown on the analyzers as the sensitivity curve and reference line.

EQ detection bands

In EQ mode, each branch has three detection bands: a low shelf, a bell, and a high shelf. Each band has a frequency, a sensitivity, and a Q, shown on the analyzer. Use them to detect, for example, kick attack in the lows and cymbal sustain in the highs separately.

Band value	Range · default	What it does
Frequency	20 Hz to 20 kHz · Low 100 Hz, Mid 1 kHz, High 8 kHz	Where the band sits.
Sensitivity	0 to 1 · 0.5	How readily the band detects.
Q	0.1 to 10 · 1	Width of the band. Higher Q is narrower.

Monitor

Control	Options · default	What it does
MONITOR	NORMAL, ATTACK, SUSTAIN · NORMAL	What you listen to. NORMAL is both shaper branches; ATTACK and SUSTAIN apply only that branch. All three run through the same output chain.
DELTA	Off, On · Off	Hear only the difference against the latency-aligned input. ATTACK with DELTA is what the attack branch adds.

MONITOR and DELTA are saved with the session. Return to NORMAL with DELTA off before you export.

Output chain

Control	Range · default	What it does
SATURATION		
On/off lamp	Off, On · Off	Switches the saturation stage on or off.
Character	Soft, Punch, Warm, Digital, Fold, Hard · Soft	Saturation character.
DRIVE	0 to 36 dB · 0 dB	How hard the signal is pushed into the selected character.
Shaper Link	Off, On · Off	Makes saturation intensity follow the attack and sustain detector activity.
CLIPPER		
On/off lamp	Off, On · Off	Switches the soft clipper on or off.
THRESHOLD	-20 to 0 dB · -0.7 dB	Level where the soft clipper starts rounding peaks off.
LIMITER		
On/off lamp	Off, On · Off	Switches the output limiter on or off.
CEILING	-24 to 0 dB · -0.1 dB	The level the limiter does not let the output pass.
LOOKAHEAD	0 to 20 ms · 0 ms	Limiter lookahead. It only has an effect while the LIMITER is on.
QUALITY AND GAIN		
QUALITY	x1, x2, x4, x8 · x1	Oversampling around saturation and clipping. x1 has the lowest CPU and latency; x8 the highest quality and CPU.
INPUT	-24 to +24 dB · 0 dB	Level going into the detector. Raise it to make quiet material trigger more bands.
OUTPUT	-24 to +24 dB · 0 dB	Final output level, after saturation, clipper, and limiter.
BYPASS	Off, On · Off	Bypasses all processing and reports zero plug-in latency while active.

The spectral analysis adds latency, which the plug-in reports to the host. Oversampling and limiter lookahead add more. BYPASS reports zero latency while it is on.

05 Presets

The preset browser holds factory and user presets. Selecting one replaces every parameter as a single undo step. The arrows step through the bank; hold one to keep stepping. Save stores the current settings as a user preset, and LOAD USER PRESET... opens a saved file. After a change, the name shows (Custom).

Group	Presets
—	Init
Drums	Snap Kick, Room Kill, Big Room, Top End Crack, Sub Punch, Parallel Slam

Group	Presets
Bass	Finger Definition, Sub Hold, Amp Grit
Guitar & Keys	Pick Attack, Soft Strum, Piano Bloom, Metallic Edge
Vocal	Consonant Control, Breath Forward, Vocal Presence
Bus & Master	Glue Tighten, Mix Punch, Open The Mix, Master Safety
Spectral FX	Reverse Swell, Gated Spikes, Frozen Tail, Broken Transients

06 Recipes

Starting points for common jobs. Adjust them by ear.

Punchier kick

More click and impact without more level.

Preset	Snap Kick
ATTACK GAIN	+3 to +8 dB
Attack RELEASE	20 to 60 ms

- 1 Monitor ATTACK with DELTA to hear only what is added.
- 2 Match OUTPUT before comparing with bypass.

Drier drum room

Shorter room tails on overheads or a drum bus.

Preset	Room Kill
SUSTAIN GAIN	-3 to -9 dB
Sustain RELEASE	Longer for smoother results

- 1 Check in NORMAL that the cymbals still ring naturally.

Clearer consonants

Speech or vocals that need definition.

Preset	Consonant Control
SHAPER	EQ
Bands	Detect in the mid and high bands

- 1 Use only the mid and high band sensitivity so the low end is not pumped.

Safe bus punch

Transient emphasis on a bus with peak safety.

Preset	Mix Punch or Master Safety
LIMITER	On
CEILING	-0.1 to -1 dB

- 1 Keep the shaping small; the limiter only catches peaks.

07 Tips and troubleshooting

Symptom	What to check
I only hear clicks or a thin sound.	MONITOR is on ATTACK or DELTA is on. Return to NORMAL with DELTA off.
Quiet parts are not shaped.	Raise INPUT so quiet material triggers more bands, or raise the sensitivity.
The low end pumps.	Use EQ mode and lower the low band sensitivity, or lengthen the releases.
Transients sound harsh after saturation.	Lower DRIVE, choose a softer character, or raise QUALITY to reduce aliasing.

Symptom	What to check
LOOKAHEAD does nothing.	It only works while the LIMITER is on.
The track is out of time with others.	Make sure host latency compensation is on. Oversampling and lookahead add latency.
A preset does not load.	The file must be a TB Spectral Transient Shaper preset made for a compatible version.

08 Parameter reference

Panel label	Host parameter	Range	Default
SHAPER			
SHAPER	Mode	Simple, EQ	Simple
ATTACK GAIN	Attack Gain	−24 to +24 dB	+4 dB
SUSTAIN GAIN	Sustain Gain	−24 to +24 dB	−2 dB
Attack analyzer	Attack Sensitivity	0 to 1	0.5
Sustain analyzer	Sustain Sensitivity	0 to 1	0.5
RELEASE	Attack Release	1 to 2000 ms	50 ms
RELEASE	Sustain Release	1 to 2000 ms	50 ms
EQ DETECTION BANDS			
Analyzer	ATK_LOW / ATK_MID / ATK_HIGH Frequency	20 Hz to 20 kHz	100 Hz / 1 kHz / 8 kHz
Analyzer	SUS_LOW / SUS_MID / SUS_HIGH Frequency	20 Hz to 20 kHz	100 Hz / 1 kHz / 8 kHz
Analyzer	ATK_... and SUS_... Sensitivity (6)	0 to 1	0.5
Analyzer	ATK_... and SUS_... Q (6)	0.1 to 10	1
MONITOR AND GAIN			
MONITOR , DELTA	Monitor	Normal, Normal Delta, Attack, Attack Delta, Sustain, Sustain Delta	Normal
INPUT	Input Gain	−24 to +24 dB	0 dB
OUTPUT	Output Gain	−24 to +24 dB	0 dB
BYPASS	Bypass	Off, On	Off
OUTPUT CHAIN			
Saturation lamp	Saturation	Off, On	Off
Character	Saturation Mode	Soft, Punch, Warm, Digital, Fold, Hard	Soft
DRIVE	Saturation Drive	0 to 36 dB	0 dB
Shaper Link	Saturation Shaper Link	Off, On	Off
Clipper lamp	Clipper	Off, On	Off

Panel label	Host parameter	Range	Default
THRESHOLD	Clipper Threshold	−20 to 0 dB	−0.7 dB
Limiter lamp	Limiter	Off, On	Off
CEILING	Limiter Threshold	−24 to 0 dB	−0.1 dB
LOOKAHEAD	Lookahead	0 to 20 ms	0 ms
QUALITY	Oversampling	x1, x2, x4, x8	x1

09 Version history

Version	Changes
1.4.1 · 2026-08-21	Brought the latest attack and sustain graphs and the current spectral processing build to Windows VST3, and aligned the catalog, VST3, and installer versions at 1.4.1.
1.4.0 · 2026-08-02	The EQ-mode analyzer and the processing now use the same sensitivity curve and reference line. Delta behaviour and latency are consistent across monitor sources, and the analyzer draws more efficiently.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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